# The Heat Is On!

Fall 2023 ARML Power Contest

## The Background

Often, in races such as at track or swim meets, there are more competitors than can compete at one time. For instance, there may be 20 racers but the track only has 8 lanes. The race is accomplished by having several smaller races called *heats*. Several racers will race in the first heat, then several more in the second heat, and so on until all the racers have raced at least once. These are called the *preliminary heats*. Then, the fastest racers will race together in the final championship heat.

The fastest racers that race in the championship will usually be chosen from the preliminary heats by using a stopwatch, and the fastest racers will be chosen regardless of how they did in their particular heat. For example, if the championship heat will have six racers, those six could all come from the same preliminary heat if those six had the fastest times among all racers in all preliminary heats. It is also possible that racer $A$ might beat racer $B$ in a preliminary heat, but in the finals $B$ might still beat $A$ (perhaps $A$ became fatigued or $B$ was "holding back" in the preliminary heat). This Power Problem will simplify the situation by not allowing for these considerations. The following assumptions will be made about the racers:

- Each racer always races with the same constant speed. If racer $A$ is faster than racer $B$ (i.e., $A$ has a larger speed than $B$) then any time $A$ and $B$ are in the same heat, $A$ will beat $B$ in that heat.
- No two racers have the same speed. There are never any ties.
- Speed comparisons are transitive. If racer $A$ is faster than racer $B$, and racer $B$ is faster than racer $C$, then racer $A$ is faster than racer $C$.
- Any two racers can be in a heat together, so that for any two racers, one of them will be faster than the other.

These assumptions imply that the racers can be put in a specific order: fastest, second place, and so on.

- There is no way to "time" the racers or otherwise determine their speeds. Only the relative speeds of racers (that is, which one is faster and which is slower) can be determined by racing them.

The last assumption means that to tell which of $A$ or $B$ is faster they must either be in the same heat together, or there is a chain of racers $C_1$, $C_2$, …, $C_n$ so that $A$ beats $C_1$ in some heat, $C_1$ beats $C_2$ in some heat, …, $C_{n-1}$ beats $C_n$ in some heat, and $C_n$ beats $B$ in some heat. These heats don't have to take place in any particular order, though; only the results of each heat are important for ranking the racers.

The goal of this problem set is to investigate how many heats are needed to determine the fastest racers. To this end, make the

**Definition:** $H(n, p, r)$ is the minimum number of heats required to determine the fastest $p$ racers (or "places") from a set of $n$ racers, where they race in heats of $r$ racers at a time.

You, as the race organizer, can choose any racers you want for each heat. You will make these decisions based on the results of earlier heats—if you are looking for the fastest $p = 3$ racers and racer $Z$ finished fifth in some heat, there is hardly any benefit to having $Z$ race again if there are other racers who are still in contention for the top 3 places! $H(n, p, r)$ is the minimum number of heats needed to ensure that you can find the fastest $p$ racers regardless of the outcomes of the heats.

**Example 1:** $H(3, 3, 2) = 3$. Here we have three racers and we want to find out who is fastest, middle, and slowest (3 places). We can race them two-at-a-time. So race $A$ against $B$, and then race $A$ against $C$. If $A$ beats $B$ and then $C$ beats $A$, we have all three places determined. But if $A$ beats both $B$ and $C$, we don't yet know which of $B$ or $C$ is faster, and must have a third race.

**Example 2:** $H(4, 2, 3) = 2$. There are four racers, and we want to determine the two fastest while racing them three-at-a-time. Race any three of them. The third place racer is eliminated. Then race the top two against the remaining racer, to see which are the two fastest.

Floor and ceiling notation will be used freely in this problem: $\lfloor x \rfloor$ is the greatest integer that is not greater $x$ and $\lceil x \rceil$ is the least integer that is not less than $x$. Additionally, the notation $\lg(x)$ may be used as shorthand for $\log_2(x)$.

## The Problems

**There are 40 points available on this contest. The numbers at the end of each problem are the number of points that problem is worth. These are also indicated on the cover sheet, which should be the front page of your solution packet. Please make sure to write on only one side of the answer sheets.**

1. Explain why $H(n, p, n) = 1$ for any $n$ and $p \leq n$. [1 pt.]

2. Explain why $H(n, n, r) = H(n, n-1, r)$ for any choices of $n$ and $r$. [1 pt.]

3. Explain why $H(n, n, n-1) = 3$ for any $n$. [2 pts.]

4. Show that $H(n, 1, r) = \left\lceil \dfrac{n-1}{r-1} \right\rceil$ for any $n$ and $r \leq n$. [3 pts.]

5. Show that $H(5, 5, 3) = 4$. [2 pts.]

6. Show that if $n \geq 6$ then $H(n, n, n-2) = 3$. [2 pts.]

7. Generalize the previous two problems. For each $k$ there is a minimum value $M$ so that $H(n, n, n-k) = 3$ for all $n \geq M$. For smaller values of $n$, the number, $n-k$, of racers in each heat is too small to determine all $n$ places in only three heats (for instance, as the previous two problems showed, for $k = 2$, $M = 6$ will work, but for $n = 5$ *four* races are required). Compute the value of $M$ in terms of $k$. Prove that your formula is correct. [4 pts.]

8. Compute, with explanation, $H(6, 3, 3)$. [3 pts.]

9. One possible strategy for finding the fastest racer is "king of the hill." Race $r$ racers, and whoever wins that heat races against $r-1$ new challengers, the winner continuing to the next heat until everyone has raced at least once. Use this strategy to show that $H(n, 2, r) \leq \left\lceil \dfrac{n-1}{r-1} \right\rceil + \left\lceil \dfrac{n-1}{(r-1)^2} \right\rceil$. [3 pts.]

10. **Example 2** in the background material explained why $H(4, 2, 3) = 2$. Find, with proof, the set of all triples $(n, p, r)$ for which $H(n, p, r) = 2$. [4 pts.]

11. Let $n = a + b$ for positive integers $a, b \geq 2$. Show that $H(n, 2, 2) \leq H(a, 2, 2) + H(b, 2, 2) + 2$. [2 pts.]

12. Use the previous problem to show that $H(n, 2, 2) \leq \lceil 3n/2 \rceil - 2$. [2 pts.]

The true value of $H(n, 2, 2)$ is actually $n - 2 + \lceil \lg(n) \rceil$. This will be proven in the next few problems. For the next three questions, assume that $r = 2$ so that all heats involve just two racers "going head-to-head."

Racing two racers at a time, using the king of the hill strategy to find the fastest racer, it is possible that the final challenger wins the last race, and only has to win one race to prove they are the fastest racer. It is also possible that the fastest racer is one of the two to compete in the first heat and then has to beat $n-2$ more challengers, requiring $n-1$ total wins to prove their supremacy.

A different way of scheduling the racers into heats is a *single-elimination bracket tournament.* This tournament is divided into rounds; each racer must race at most once in each round. For the first round of racing each racer is paired with an opponent and they race. The loser is eliminated from the competition and the winner goes on to the next round. If there are an odd number of racers, one gets a "bye" and is moved into the next round without having to race. Then the procedure is repeated in each round, eliminating half the racers each round until a single winner emerges.

13. As a function of $n$, compute the maximum possible number of heats that a racer must win in a single-elimination bracket tournament to prove they are the fastest racer. [2 pts.]

14. Show that there is no way of scheduling racers into heats so that the eventual winner is guaranteed to have to compete fewer times than in a single-elimination bracket tournament. [2 pts.]

15. Prove that $H(n,2,2) = n-2+\lceil \lg(n) \rceil$. [2 pts.]

Assume the top $p$ places have been determined, and then another racer appears. How many extra heats will it take to determine if and where this racer belongs in those top $p$ places?

16. Prove that $H(n+1,p,r) \leq H(n,p,r) + \lceil \log_r(p) \rceil + 1$.

17. This Power Contest was inspired by a puzzle given to the author. In this puzzle, there are 25 horses that always run at the same speed, and can be raced five-at-a-time. The goal is to determine the three fastest horses, in order. Solve the puzzle—determine, with explanation, $H(25,3,5)$. [3 pts.]

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In the penultimate problem it should be noted that $\lceil \log_r(p) \rceil + 1$ is the number of *additional* races that would be necessary if the new racer hadn't been available to compete in the original heats. In general, if that racer had been part of the competition from the outset there is a more efficient strategy. As an example, $H(6,2,2) = 7$ and $H(7,2,2) = 8$ which is less than $H(6,2,2) + \lceil \log_2(2) \rceil + 1 = 7+1+1 = 9$. This shows that it is *not* the most efficient strategy to completely rank a subset of the racers and then add more competitors in; it is more efficient to allow as many of the racers to compete at each stage as possible and not to requires a racer to race again until all others have been in at least the same number of races.

The exact values of the function $H(n,p,r)$ are unknown. This problem is a specific form of *partial sorting* where more than two items may be compared at a time. Asymptotic bounds are known (both upper and lower). A simple web search on sorting and partial sorting algorithms will lead you to a wealth of information about this topic.